



NPTEL Online Certification Course
Applied Accelerated Artificial Architecture
Assignment 6 : Detailed Solution
Indian Institute of Technology Guwahati

Question1. What primary training objective is used during the pretraining stage of Large Language Models (LLMs)?

Options:

- A. Masked token prediction
- B. Next-token prediction
- C. Contrastive sequence matching
- D. Direct preference optimization

Correct Answer: B. Next-token prediction

Explanation:

During pretraining, LLMs learn language representations by predicting the probability distribution of the next token given preceding context.

This self-supervised autoregressive objective enables models to learn from massive amounts of unlabeled text without human annotation.

Question2. Which adaptation approach modifies external retrieved context rather than updating any internal model parameters?

Options:

- A. Parameter-Efficient Fine-Tuning (PEFT)
- B. Retrieval-Augmented Generation (RAG)
- C. Supervised Fine-Tuning (SFT)
- D. Low-Rank Adaptation (LoRA)

Correct Answer: B. Retrieval-Augmented Generation (RAG)

Explanation:

RAG fetches relevant document passages from external sources and injects them directly into the input prompt at inference time.

Because it operates solely by augmenting the prompt context, it expands model knowledge without modifying any internal model weights.

Question3. Beyond model parameters, which component often contributes significantly to memory overhead during full model fine-tuning?

Options:

- A. Static input token IDs
- B. Tokenizer vocabulary lookup tables
- C. Fixed positional embedding matrices
- D. Optimizer states and gradients

Correct Answer: D. Optimizer states and gradients

Explanation:

Full model fine-tuning requires tracking first and second momentum states along with weight gradients for every single trainable parameter.

These optimizer states and gradients combined consume several times more GPU memory than the base model parameters themselves.

Question4. What primary operation converts input text into subword units before fed into an LLM embedding layer?

Options:

- A. Tokenization
- B. Normalization
- C. Quantization
- D. De-noising

Correct Answer: A. Tokenization

Explanation:

Tokenization decomposes raw input text strings into subword units or characters based on a predefined vocabulary.

This subword tokens are mapped to discrete numerical IDs that index directly into the model's dense embedding matrix.

Question5. What is the primary benefit of utilizing Brain Float 16 (BF16) over conventional Floating Point 16 (FP16) when training large transformer models?

Options:

- A. It uses fewer total bits to store numerical values.
- B. It has a wider exponent range.
- C. It automatically applies gradient scaling routines.
- D. It can safely downcast the final output tensors.

Correct Answer: B. It has a wider exponent range.

Explanation: BF16 maintains an 8-bit exponent field matching standard FP32, effectively preventing numerical underflow and overflow during training.

This expanded dynamic range removes the need for complex loss scaling workarounds required by FP16's narrow 5-bit exponent.

Question6. How can the numerical issue of naive softmax calculations blowing up into "Not a Number" (NaN) or infinity be successfully stabilized?

Options:

- A. By manually calling a blanket model.half() function.
- B. By switching the target system architecture to a legacy GPU.
- C. By executing the backward pass within an autocast region.
- D. By subtracting the maximum value from the logits before softmax.

Correct Answer: D. By subtracting the maximum value from the logits before softmax.

Explanation:

Subtracting the maximum logit value from all input logits ensures that the maximum exponent evaluated during softmax is zero. This shift-invariant mathematical trick prevents numeric overflow during exponentiation without changing the final output probabilities.

Question7. What occurs when tiny gradients flush directly to zero during the training process, and what symptom describes this state?

Options:

- A. Underflow, causing the learning process to stall.
- B. Underflow, causing the loss function to hit infinity.
- C. Overflow, causing the parameters to immediately vanish.
- D. Overflow, causing logits to blow up before the softmax

Correct Answer: A. Underflow, causing the learning process to stall.

Explanation:

Underflow occurs when tiny gradient values fall below the representable numerical precision limit and round down to zero. Without non-zero gradient values to update model weights, backpropagation produces no parameter changes and training stalls.

Question8. Which of the following components has the same shape as the model weights when training an LLM?

Options:

- A. Optimizer states

- B. Gradients
- C. Activations
- D. Framework overhead

Correct Answer: B. Gradients

Explanation:

Gradients represent partial derivatives of the scalar loss function computed with respect to each individual model parameter. Because every trainable weight requires a corresponding gradient value, gradient tensors share an exact 1:1 structural shape with weights.

Question9. What is the primary trade-off involved when implementing gradient checkpointing in LLM training?

Options:

- A. Trading compute for memory
- B. Trading precision for speed
- C. Trading accuracy for memory
- D. Trading throughput for precision

Correct Answer: A. Trading compute for memory

Explanation:

Gradient checkpointing drastically reduces memory usage by discarding intermediate activation tensors during the forward pass.

It then recomputes these missing activations on-demand during the backward pass, increasing total FLOPs in exchange for lower VRAM footprint.

Question10. What happens to a weight update step in dynamic loss scaling if an 'inf' or 'nan' value is detected in the gradients?

Options:

- A. The update step is skipped entirely.
- B. The update step runs with a fixed scale.
- C. Only the update of corrupted weights are skipped.
- D. The model falls back to full precision.

Correct Answer: A. The update step is skipped entirely.

Explanation:

An 'inf' or 'nan' value indicates numeric overflow during backpropagation, which would permanently corrupt model parameters if applied.

The framework skips the parameter update step, reduces the scale factor for subsequent batches, and resumes training safely.

Question11. Which parameter-efficient fine-tuning technique is specifically recommended when weight updates are necessary but the memory or GPU budget is highly limited?

Options:

- A. Supervised Fine-Tuning (SFT)
- B. Continued Pre-training (CPT)
- C. Low-Order Rank Adaptation (LoRA)
- D. Zero-Shot Generation

Correct Answer: C. Low-Order Rank Adaptation (LoRA)

Explanation:

LoRA freezes pretrained model weights and introduces small, trainable low-rank decomposition matrices into transformer layers.

This reduces trainable parameters and optimizer memory by orders of magnitude, making fine-tuning feasible on limited GPU hardware.

Question12. What major limitation was observed when testing the base, pre-trained GPT-2 model using a zero-shot generation prompt?

Options:

- A. It did not know how to follow the specified instruction format.
- B. It failed to generate grammatically correct English sentences.
- C. It was unable to process abstract concepts like networks.
- D. It completely refused to output text due to policy constraints.

Correct Answer: A. It did not know how to follow the specified instruction format.

Explanation:

Base pretrained models are optimized purely for raw autoregressive text continuation rather than task or instruction execution. Without instruction fine-tuning, zero-shot prompts often cause the model to repeat or extend the prompt rather than output the answer format.

Question13. When setting up a model's weights to be frozen so that the backward pass does not update them during training, which configuration must be applied?

Options:

- A. `requires_grad = False`
- B. `requires_grad = True`
- C. `double_quantization = True`
- D. `optimizer_state = True`

Correct Answer: A. requires_grad = False

Explanation:

Setting a tensor's requires_grad attribute to False excludes it from autograd computation graph construction. This prevents autograd from computing gradients for those parameters during backpropagation, leaving the weights frozen.

Question14. During the practical implementation of LoRA, how are the weight matrices A and B typically initialized?

Options:

- A. Matrix A is initialized with zeros, and Matrix B is initialized from a Gaussian distribution.
- B. Matrix A is initialized from a Gaussian distribution, and Matrix B is initialized with zeros.
- C. Both Matrix A and Matrix B are initialized with zeros.
- D. Both Matrix A and Matrix B are drawn from a uniform float distribution.

Correct Answer: B. Matrix A is initialized from a Gaussian distribution, and Matrix B is initialized with zeros.

Explanation:

Initializing matrix B to zeros ensures that the rank update product $\Delta W = B \cdot A$ equals zero at the start of training. This guarantees that the model's baseline behavior matches original pretrained weights prior to fine-tuning updates.

Question15. If a task requires parameter updates, why is PEFT typically recommended as the first adaptation strategy instead of full fine-tuning?

Options:

- A. It optimizes the entire model framework from scratch to eliminate noise.
- B. It completely avoids the backward pass workflow during training.
- C. It adds and trains only a few parameters to test baseline performance.
- D. It guarantees higher structural throughput by using 4-bit normal floats.

Correct Answer: C. It adds and trains only a few parameters to test baseline performance.

Explanation: PEFT trains only a tiny fraction of the total model parameters, dramatically lowering compute time, VRAM, and storage requirements.

This allows rapid baseline evaluation and task adaptation before committing resources to full model fine-tuning.